

Dalron J. Robertson, M.S.

Research Scientist, Pharm.D./MBA Candidate

dalronj.robertson@gmail.com | Chicago, IL | <https://dalronjrobertson.com> | <https://nasresearch.bio>
<https://www.linkedin.com/in/dalronjrobertson/> | <https://github.com/AGuyNamedDJ>

Education

Chicago State University – *Doctor of Pharmacy, Expected May '28*

Mississippi College – *MBA, Expected May '28*

Mississippi College – *M.S., Biological Sciences*

University of Illinois, Chicago – *Software Engineering Certificate*

Jackson State University – *B.S., Biology*

Technical Skills

AI & Large Language Models: Deepseek | FAISS | Gemini | LLaMA | LoRA | OpenAI | TinyLlama | TxGemma

Bioinformatics & Molecular Modeling: AlphaFold | ChimeraX | SwissTargetPrediction | Molinspiration | RDKit | UniProt

DataScience & Machine Learning: Tensorflow | Scikit-Learn | Pandas | NumPy | PyTorch

Languages: Javascript | Python | Pinescript | Java | C | C++ | PostgreSQL | MySQL

Tools & Technologies: Git | AWS | ISOLIDE | | Schrödinger | jest | DocuTrack | Framework LTC | UiPath

Web Development: Node.js | React.js | Express.js | FastAPI | Flask | HTML | CSS | AJAX | DOM | jQuery

Experience

UChicago Medicine, Hyde Park, Chicago, IL

Pharmacy Intern — Meds 2 Beds | March 2026 -

- Support hospital transitions of care by preparing and delivering discharge medications to patient bedside and providing medication counseling prior to discharge
- Educate patients and caregivers on newly initiated and modified therapies, including dosing, adverse effects, and adherence considerations.
- Collaborate with physicians, nurses, pharmacists, and care coordinators to support medication reconciliation and discharge planning.
- Assist with outpatient pharmacy operations to ensure continuity of therapy following hospitalization.

Walgreens, Chicago, IL

Pharmacy Intern | Jan 2026 -

- Support pharmacists in medication verification, dispensing workflows, and clinical decision-making processes in accordance with state and federal pharmacy regulations.
- Interpret patient profiles, medication histories, and prescription data within pharmacy systems to ensure accuracy, appropriateness, and regulatory compliance prior to dispensing.
- Contribute to patient care services within intern scope, including medication counseling support and preventive health initiatives, while upholding controlled substance and Good Faith Dispensing standards.

Astellas Pharmaceuticals, Northbrook, IL

Mentee - Bridge to Industry Intensive Program | Oct 2025 -

- Selected participant in a professional development partnership between Chicago State University College of Pharmacy and Astellas Pharma Global Development.
- Engage in one-on-one mentorship with Astellas executives, gaining exposure to Quality, Research, and Regulatory leadership.
- Participate in seminars and applied learning sessions focused on pharmaceutical innovation, compliance, and global development pathways.

Mississippi College, Clinton, MS

Lead Graduate Researcher | Dec 2024 - Jul 2025

- Applied AI-driven literature mining and bioinformatics to extract and synthesize bioactivity, phytochemical, and pharmacological data from large biomedical databases.
- Built and analyzed a curated library of *Nigella sativa* constituents, performing molecular descriptor analysis, drug-likeness screening, and lead prioritization.
- Identified and modeled protein targets using SwissTargetPrediction and AlphaFold, and conducted molecular docking and visualization to assess ligand-target interactions.

Scale AI, San Francisco, CA

AI Software Engineer | Sept 2023 - Jan 2025

- Collaborated with Google teams to improve Gemini and Google Search models by developing high-quality training datasets that enhanced accuracy, relevance, and performance.
- Performed side-by-side model evaluations and red teaming exercises to identify weaknesses, mitigate misuse risks, and inform real-time model improvement.
- Analyzed large-scale model outputs and built synthetic data pipelines to improve temporal and relational reasoning, contributing to measurable gains in model precision.

Medication Management Partners, Crestwood, IL

Pharmacy Informatics & Automation, Pharmacy Intern | Aug 2020 - Apr 2022

CVS Health, Chicago, IL

Pharmacy Intern | Jul 2015 - Aug 2020

Research & Development Projects

NaS Knowledge Model | June 2025 | <https://github.com/NaS-Research/knowledge-model>

Engineered a biomedical retrieval-augmented LLM based on Google Deepmind's TxGemma 2B. The model utilizes LoRA fine-tuning to produce literature-supported responses and autonomously updates nightly with newly ingested research data.

Key Achievements & Technical Highlights:

- Reduced model training from multiple days to less than four hours using targeted LoRA adapters (error reduced from 5.16 → 2.23 within ~300 steps).
- Increased retrieval accuracy through 1,024-dimensional BGE-large embeddings and a two-stage retrieval strategy.
- Automated daily ingestion, embedding, and indexing of biomedical data, eliminating manual intervention.
- Implemented rigorous context validation, decreasing unsupported claims by approximately ~34%.

Phytochemical Bioactivity Research | Jan 2025 - May 2025 | <https://bit.ly/DJR-PhytochemicalResearch>

Conducted computational and experimental research on ethanol-derived *Nigella sativa* extracts using bioinformatics, molecular modeling, and AI-driven analysis. Presented findings at symposium; recognized by Provost for research excellence.

- **Phase 1: Literature Mining & Data Collection**
 - Leveraged Deep Research methodologies (OpenAI's LLMs) to rapidly extract and synthesize bioactivity, pharmacological, and chemical data, from PubMed, PubChem, and UniProt databases.
- **Phase 2: Compound Library & Chemical Profiling**
 - Developed a comprehensive chemical library of *N. sativa* phytochemicals, conducting detailed molecular descriptor calculations, ADME/Tox assessments, and archiving structural data for computational analyses.
- **Phase 3: Target Identification & Structure Modeling**
 - Identified protein targets via SwissTargetPrediction; validated through extensive literature reviews. Generated 3D protein models using AlphaFold; developed visualizations with ChimeraX and Blender.
- **Phase 4: Virtual Screening & Molecular Docking (Current Phase)**
 - Performing molecular docking simulations to predict binding affinities and interaction dynamics between prioritized phytochemicals and protein targets.
 - **Teaching & Mentorship:** Trained and mentored 16 undergraduate and graduate researchers on the complete computational pipeline, including software and analytical methods, significantly enhancing research productivity.